통계열역학 (2014년 1학기) 중간고사 해답

(총 4문제, 120점)

1. **(총 30점, 각 5점)**

   (a) 화학퍼텐셜

   (b) 에르고드 가설

   (c) 열탕

   (d) 자발적 과정

   (e) 자연변수

   (f) 열역학 제3법칙

2. **(총 30점)**

   (a) **(5점)**

$$\begin{aligned} Q(2,V,T) &= \sum_{\sigma} \Omega(E_\sigma) e^{-\beta E_\sigma} \\ &= \underbrace{(V-1)e^{\beta\epsilon}}_{\text{이량체에 대한 분자분배함수}} + \underbrace{\left(\frac{V}{2}-1\right)(V-1)}_{\text{단량체에 대한 분자분배함수}} \end{aligned} \tag{1}$$

   (b) **(5점)** (1) 식으로부터 Helmholtz 자유에너지 $A(2,V,T)$는 다음과 같이 계산되므로

$$\begin{aligned} A &= -k_B T \ln Q \\ &= -k_B T \ln \left[(V-1)e^{\beta\epsilon} + \left(\frac{V}{2}-1\right)(V-1) \right]. \end{aligned}$$

   평균에너지는 다음과 같이 구할 수 있다.

$$\begin{aligned} \langle E \rangle &= k_B T^2 \left(\frac{\partial \ln Q}{\partial T} \right)_{N,V} \\ &= k_B T^2 \frac{\partial}{\partial T} \ln \left[(V-1)e^{\epsilon/k_B T} + \left(\frac{V}{2}-1\right)(V-1) \right] \\ &= k_B T^2 \frac{-(V-1)\frac{\epsilon}{k_B T^2} e^{\epsilon/k_B T}}{(V-1)e^{\epsilon/k_B T} + \big(V/2-1\big)(V-1)} \\ &= \frac{-\epsilon(V-1)e^{\epsilon/k_B T}}{(V-1)e^{\epsilon/k_B T} + \big(V/2-1\big)(V-1)} \\ &= \frac{-\epsilon e^{\epsilon/k_B T}}{e^{\epsilon/k_B T} + \big(V/2-1\big)} \end{aligned}$$

   (c) **(10점)** 낮은 에너지 극한에서

$$\begin{aligned} \lim_{T\to 0} \langle E \rangle &= \lim_{T\to 0} \frac{-\epsilon e^{\epsilon/k_B T}}{e^{\epsilon/k_B T} + \big(V/2-1\big)} \\ &\approx -\epsilon \end{aligned}$$

높은 에너지 극한에서

$$\begin{aligned}\lim_{T\to\infty}\langle E\rangle &= {}_{T\to\infty}\frac{-\epsilon e^{\epsilon/k_BT}}{e^{\epsilon/k_BT}+\big(V/2-1\big)}\\ &\approx -\frac{2\epsilon}{V}\end{aligned}$$

(d) **(10 점)** 압력과 정준분배함수의 관계식으로부터

$$\begin{aligned}P &= k_BT\left(\frac{\partial \ln Q}{\partial V}\right)_T\\ &= k_BT\frac{\partial}{\partial V}\ln\left[(V-1)e^{\epsilon/k_BT}+\left(\frac{V}{2}-1\right)(V-1)\right]\\ &= k_BT\frac{e^{\epsilon/k_BT}+V-3/2}{(V-1)e^{\epsilon/k_BT}+\big(V/2-1\big)(V-1)}\end{aligned}$$

3. **(총 30 점, 각 10 점)**

(a) 분자들이 서로 구별가능하다고 할 때 정준분배함수, $Q_{\rm rot}(N,T)$ 는

$$Q_{\rm rot}(N,T)=q_{\rm rot}(T)^N=\left[\frac{\pi^{1/2}}{\sigma}\left(\frac{T^3}{\Theta_{\rm rot,a}\Theta_{\rm rot,b}\Theta_{\rm rot,c}}\right)^{1/2}\right]^N$$

(b) 평균에너지 $\langle E\rangle$ 는

$$\begin{aligned}\langle E\rangle &= k_BT^2\left(\frac{\partial \ln Q}{\partial T}\right)_{N,V}\\ &= k_BT^2\frac{\partial}{\partial T}\ln\left[\frac{\pi^{1/2}}{\sigma}\left(\frac{T^3}{\Theta_{\rm rot,a}\Theta_{\rm rot,b}\Theta_{\rm rot,c}}\right)^{1/2}\right]^N\\ &= k_BT^2N\frac{\partial}{\partial T}\left[\frac{3}{2}\ln T+\ln\left\{\frac{\pi^{1/2}}{\sigma}\left(\frac{1}{\Theta_{\rm rot,a}\Theta_{\rm rot,b}\Theta_{\rm rot,c}}\right)^{1/2}\right\}\right]\\ &= \frac{3}{2}Nk_BT\end{aligned}$$

(c)

$$\bar{C}_V=\left(\frac{\partial\langle E\rangle}{\partial T}\right)_{N_A,V}=\frac{\partial}{\partial T}\left(\frac{3}{2}N_Ak_BT\right)=\frac{3}{2}N_Ak_B=\frac{3}{2}R \qquad (2)$$

4. **(총 30 점, 각 10 점)**

(a) (단, $\epsilon_1<\epsilon_2$)

$$\begin{aligned}p_1 &= \frac{e^{-\beta\epsilon_1}}{e^{-\beta\epsilon_1}+e^{-\beta\epsilon_2}}=\frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT}+e^{-\epsilon_2/k_BT}}\\ p_2 &= \frac{e^{-\beta\epsilon_2}}{e^{-\beta\epsilon_1}+e^{-\beta\epsilon_2}}=\frac{e^{-\epsilon_2/k_BT}}{e^{-\epsilon_1/k_BT}+e^{-\epsilon_2/k_BT}}\end{aligned}$$

(b) i. $T = 0$일 때, p_1은

$$\begin{aligned}\lim_{T\to 0} p_1 &= \lim_{T\to 0} \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT} + e^{-\epsilon_2/k_BT}} \\ &= \lim_{T\to 0} \frac{1}{1 + e^{-(\epsilon_2-\epsilon_1)/k_BT}} \qquad \text{(양변을 } e^{-\epsilon_1/k_BT} \text{로 나눔)}\end{aligned}$$

$\epsilon_1 < \epsilon_2$ 0이므로

$$\boxed{\lim_{T\to 0} \frac{1}{1 + e^{-(\epsilon_2-\epsilon_1)/k_BT}} = 1}$$

한편, p_2는

$$\begin{aligned}\lim_{T\to 0} p_2 &= \lim_{T\to 0} \frac{e^{-\epsilon_2/k_BT}}{e^{-\epsilon_2/k_BT} + e^{-\epsilon_2/k_BT}} \\ &= \lim_{T\to 0} \frac{1}{e^{-(\epsilon_1-\epsilon_2)/k_BT} + 1} \qquad \text{(양변을 } e^{-\epsilon_2/k_BT} \text{로 나눔)}\end{aligned}$$

$\epsilon_1 < \epsilon_2$ 0이므로

$$\boxed{\lim_{T\to 0} \frac{1}{e^{-(\epsilon_1-\epsilon_2)/k_BT} + 1} = 0}$$

ii. $T = \infty$ 일 때, p_1은

$$\lim_{T\to\infty} p_1 = \lim_{T\to\infty} \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT} + e^{-\epsilon_2/k_BT}}$$

$\lim_{T\to\infty} e^{-\epsilon_1/k_BT} = 1$, 그리고 $\lim_{T\to\infty} e^{-\epsilon_2/k_BT} = 1$ 0이므로

$$\boxed{\lim_{T\to\infty} p_1 = \lim_{T\to\infty} \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT} + e^{-\epsilon_2/k_BT}} = \frac{1}{2}}$$

한편, p_2는

$$\lim_{T\to\infty} p_2 = \lim_{T\to\infty} \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT} + e^{-\epsilon_2/k_BT}}$$

$\lim_{T\to\infty} e^{-\epsilon_1/k_BT} = 1$, 그리고 $\lim_{T\to\infty} e^{-\epsilon_2/k_BT} = 1$ 0이므로

$$\boxed{\lim_{T\to\infty} p_2 = \lim_{T\to\infty} \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT} + e^{-\epsilon_2/k_BT}} = \frac{1}{2}}$$

(c) (a)에서

$$
\begin{aligned}
p_1 &= \frac{e^{-\epsilon_1/k_BT}}{e^{-\epsilon_1/k_BT}+e^{-\epsilon_2/k_BT}} \\
&= \frac{1}{1+e^{-(\epsilon_2-\epsilon_1)/k_BT}} \qquad (\text{양변을 } e^{-\epsilon_1/k_BT} \text{ 로 나눔}) \\
&= \frac{1}{1+e^{-\Delta\epsilon/k_BT}} \qquad (\Delta\epsilon = \epsilon_2 - \epsilon_1 \text{ 이므로}) \\
&= \frac{1}{1+e^{-1}} = 0.731 \qquad (\Delta\epsilon = k_BT \text{ 이므로})
\end{aligned}
$$

$$
\begin{aligned}
p_2 &= \frac{e^{-\epsilon_2/k_BT}}{e^{-\epsilon_1/k_BT}+e^{-\epsilon_2/k_BT}} \\
&= \frac{1}{e^{-(\epsilon_1-\epsilon_2)/k_BT}+1} \qquad (\text{양변을 } e^{-\epsilon_2/k_BT} \text{ 로 나눔}) \\
&= \frac{1}{e^{\Delta\epsilon/k_BT}+1} \qquad (\Delta\epsilon = \epsilon_2 - \epsilon_1 \text{ 이므로}) \\
&= \frac{1}{e^{1}+1} = 0.269 \qquad (\Delta\epsilon = k_BT \text{ 이므로})
\end{aligned}
$$